

Business Finance - BBA Sem 3 Notes

BBA Semester 3 — Course Code F010301TN — Detailed Notes

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Quick Revision Summary

Unit I: Introduction to Finance

(BBA Sem 3 — F010301TN — 10 Lectures)

1. Finance Functions

Finance is often described as the lifeblood of any business — every business decision, in some way, has a financial implication. The finance function refers to the set of managerial activities concerned with the **procurement (raising) and effective utilization (use) of funds** for the business.

The finance function is typically broken down into three core managerial decisions, which form the backbone of this entire subject:

1. **Investment Decision** — Deciding where to deploy the firm's funds (which assets/projects to invest in)
2. **Financing Decision** — Deciding how to raise the funds needed (what mix of debt and equity to use)
3. **Dividend Decision** — Deciding how much of the firm's profit to distribute to shareholders vs. retain for reinvestment

(These three decisions are explored in detail below, and together form the core scope of financial management — the “Various Decisions” mentioned in your syllabus.)

Beyond these three central decisions, the finance function also includes supporting activities:

- **Financial planning and forecasting** — Estimating future fund requirements
- **Financial control** — Monitoring whether funds are being used efficiently, often through budgeting and variance analysis
- **Liquidity management** — Ensuring the firm always has enough cash to meet short-term obligations
- **Profit planning** — Analyzing cost-volume-profit relationships to guide pricing and output decisions

Exam tip: “Explain the finance functions” is a common 6-mark question — center your answer on the three key decisions (Investment, Financing, Dividend) as the primary functions, then briefly mention planning, control, and liquidity management as supporting functions.

2. Objectives of Financial Management: Profit Maximization vs. Shareholder Wealth Maximization

This is one of the most conceptually important — and most frequently examined — topics in the entire subject.

Profit Maximization as an Objective

Traditionally, many argued that the objective of financial management should be to maximize the firm's profits, since profit is seen as a measure of efficiency and a necessary condition for survival and growth.

Arguments in favor:

- Profit is a straightforward, easily understood measure of business performance
- It reflects economic efficiency in using scarce resources
- Higher profits generally mean better returns and resource utilization

Limitations of Profit Maximization (why it is criticized as an objective):

- **Ignores the time value of money** — It doesn't distinguish between profit earned today versus profit earned several years later, even though money received earlier is worth more.
- **Ignores risk** — Profit maximization does not account for the riskiness of different courses of action; a firm might chase higher profit through very risky decisions that could jeopardize its long-term survival.
- **Ambiguity in the term "profit"** — Profit can be measured in different ways (short-term vs long-term, before or after tax, total profit vs. profit per share), making the objective vague and open to manipulation.
- **Ignores the quality and timing of earnings** — Doesn't account for how consistent or sustainable those profits are.

Shareholder Wealth Maximization (SWM) as an Objective

This is now widely accepted as the more appropriate objective of financial management. It states that the goal of financial management should be to **maximize the market value of the shareholders' wealth**, which is reflected in the market price of the company's equity shares.

Why SWM is considered superior to profit maximization:

- **Considers the time value of money** — Wealth is measured in

terms of cash flows, which are adjusted for timing (discounted appropriately), unlike simple accounting profit.

- **Considers risk** — SWM explicitly factors in the risk associated with the stream of expected benefits/cash flows, since riskier cash flows are discounted more heavily.
- **Precise and unambiguous** — Market price of a share is an objective, observable figure (in the case of listed companies), unlike the somewhat vague concept of “profit.”
- **Reflects long-term value creation** — Since share price reflects the market’s assessment of the firm’s future earning potential, SWM inherently pushes managers toward long-term, sustainable decision-making rather than short-term profit boosting.

Comparison Table

BASIS	PROFIT MAXIMIZATION	SHAREHOLDER WEALTH MAXIMIZATION
Focus	Short-term accounting profit	Long-term market value of shares
Time value of money	Ignored	Explicitly considered
Risk	Ignored	Explicitly considered
Precision	Ambiguous (many profit measures)	Precise (market price is observable)
Time horizon	Short-term oriented	Long-term oriented

Exam tip: “Profit maximization vs. shareholder wealth maximization” is a very high-probability 6-mark question — structure your answer with the limitations of profit maximization first, then present SWM as the superior alternative, using the comparison table above.

3. Various Decisions in Financial Management (Investment, Financing & Dividend)

These three decisions represent the complete scope of what a finance manager does, and they are deeply interlinked.

A. Investment Decision

The investment decision relates to **how the firm’s funds are invested in different assets**, so as to earn the highest possible return for a given level of risk. This decision has two components:

- **Long-term investment decision (Capital Budgeting)** —

Deciding whether to invest in long-term assets/projects (e.g., new machinery, a new factory), since these decisions involve large sums of money and long-term commitments. (This is covered in depth in Unit II.)

- **Short-term investment decision (Working Capital Management)** — Deciding how much to invest in current assets (cash, inventory, receivables) to ensure smooth day-to-day operations without tying up excessive funds. (Covered in Unit IV.)

The investment decision is often considered the **most important** of the three decisions, because a firm's overall profitability and value depend heavily on how wisely it selects and manages its assets/projects.

B. Financing Decision

The financing decision relates to **how the firm should raise the funds required for investment** — essentially, deciding on the right mix of debt and equity (referred to as the firm's "capital structure"). Key considerations:

- **Cost of different sources of finance** — Debt is generally cheaper than equity (partly due to tax benefits on interest), but excessive debt increases financial risk.
- **Risk-return trade-off** — More debt increases the potential return to equity shareholders (financial leverage) but also increases the risk of financial distress.
- **Control considerations** — Raising funds through equity may dilute existing owners' control over the company, while debt does not.

(This decision is explored in depth in Unit III — Capital Structure.)

C. Dividend Decision

The dividend decision relates to **how much of the firm's profit should be distributed to shareholders as dividends, and how much should be retained** for reinvestment in the business. Key considerations:

- Retained earnings are a low-cost source of internal financing for growth
- However, shareholders may also value a steady stream of dividend income
- The decision must balance shareholder expectations for current income against the firm's need for funds to finance future growth

(This decision is explored in depth in Unit IV — Dividend Decision.)

Exam tip: A 6-mark question may directly ask “Explain the major decisions of financial management” — structure the answer using these three decisions, explicitly naming Investment, Financing, and Dividend decisions, with 2-3 lines on each.

4. Time Value of Money (TVM)

Concept

The Time Value of Money is one of the most fundamental principles in finance. It states that **a rupee received today is worth more than a rupee received in the future**, because money available today can be invested to earn a return, growing larger over time.

Reasons money has time value:

- **Opportunity for earning/investment** — Money in hand today can be invested to earn interest/returns.
- **Inflation** — The purchasing power of money tends to decrease over time due to rising prices.
- **Risk and uncertainty** — Future cash flows are less certain than money already in hand today (a “bird in hand” is worth more than a promise of two in the future).
- **Personal preference for current consumption** — People generally prefer to consume now rather than wait, all else being equal.

Present Value (PV) and its Relationship with Discount Rate

Present Value is the current worth of a future sum of money, given a specified rate of return (called the “discount rate”). It answers the question: “How much would I need to invest today, at a given interest rate, to receive a specific amount in the future?”

The basic present value formula:

$$PV = FV / (1 + r)^n$$

Where: FV = Future Value, r = discount rate (rate of return), n = number of periods

Relationship between Present Value and Discount Rate:

- Present Value and the Discount Rate have an **inverse relationship** — as the discount rate increases, the present value of a given future sum decreases, and vice versa.
- This is because a higher discount rate implies a higher

opportunity cost of money (or higher risk), meaning less money needs to be set aside today to grow into the same future amount.

- Similarly, PV and the number of time periods (n) also have an inverse relationship — the further away a cash flow is in the future, the lower its present value (holding the discount rate constant).

Exam tip: “Explain the relationship between present value and discount rate” is a common 2-mark/6-mark question — the key point to state clearly is the **inverse relationship**, with a brief explanation of why (higher discount rate = higher opportunity cost = lower present value needed today).

Future Value (FV)

Future Value is the value of a current sum of money (or a series of payments) at a specified date in the future, assuming a certain rate of return.

$$FV = PV \times (1 + r)^n$$

5. Present Value and Future Value of an Annuity

Concept of an Annuity

An annuity is a **series of equal periodic payments (or receipts) made at fixed intervals over a specified period of time** (e.g., a fixed monthly loan installment, a fixed yearly insurance premium, or a fixed pension payout).

Types of Annuities

- **Ordinary Annuity (Annuity Regular)** — Payments occur at the **end** of each period (e.g., loan EMI payments made at month-end).
- **Annuity Due** — Payments occur at the **beginning** of each period (e.g., rent typically paid at the start of the month).

Present Value of an Annuity

The Present Value of an Annuity is the current worth of a series of equal future payments, discounted at a given interest rate. It essentially answers: “What lump sum today is equivalent to receiving this fixed series of payments in the future?”

General logic: Since each payment in the annuity occurs at a different point in time, each payment is individually discounted back to present value, and then all these present values are summed. Annuity tables/formulas simplify this into a single calculation.

Future Value of an Annuity

The Future Value of an Annuity is the accumulated value of a series of equal periodic payments/deposits at a specified future date, assuming each payment earns a given rate of return from the time it is made until the final date.

General logic: Earlier payments in the series have more time to earn compound interest, so they contribute more to the final future value than payments made later in the series.

Exam tip: Numerical problems on PV/FV of annuity are common in this unit — make sure you understand the conceptual distinction between ordinary annuity and annuity due, since this affects which formula/table is applied.

6. Valuation of Bonds

Concept

Bond valuation is the process of **determining the fair (theoretical) price of a bond**, based on the present value of all the cash flows the bond is expected to generate for its holder in the future.

A bond typically generates two types of cash flows for its holder:

1. **Periodic interest (coupon) payments** — Fixed interest payments made at regular intervals (e.g., annually or semi-annually) until maturity
2. **Repayment of the face value (principal/par value)** — Paid at maturity, at the end of the bond's life

Bond Valuation Logic

Since a bond's value is essentially the present value of its future cash flows, bond valuation involves:

- Discounting each future coupon payment back to its present value, using an appropriate discount rate (usually the market interest rate/required rate of return)
- Discounting the final face value repayment back to present value
- Summing these present values to arrive at the bond's fair value

Key relationships in bond valuation:

- **Bond price and interest rates move inversely** — When market interest rates rise, the value of existing bonds (with fixed coupon rates) falls, because their fixed payments become less attractive relative to newly issued bonds offering higher rates. When market interest rates fall, existing bond prices rise.

- **Coupon rate vs. Market/Required rate:**
 - If the coupon rate equals the market rate, the bond sells at **par value** (face value)
 - If the coupon rate is greater than the market rate, the bond sells at a **premium** (above face value)
 - If the coupon rate is less than the market rate, the bond sells at a **discount** (below face value)

Exam tip: “Explain the concept of bond valuation” and “relationship between bond price and interest rate” are both frequently tested — remember the inverse relationship and the three premium/discount/par scenarios above, as these are the most likely conceptual (non-numerical) questions.

Quick Revision Summary

TOPIC	ONE-LINE TAKEAWAY
Finance function	Procurement and effective utilization of funds — via Investment, Financing, Dividend decisions
Profit Maximization	Simple but ignores time value of money, risk, and is ambiguous
Shareholder Wealth Maximization	Superior objective — considers time value of money, risk, and is precise (market price)
Investment decision	Where to deploy funds — long-term (capital budgeting) and short-term (working capital)
Financing decision	How to raise funds — the right mix of debt and equity
Dividend decision	How much profit to distribute vs. retain
Time Value of Money	A rupee today is worth more than a rupee tomorrow (due to earning potential, inflation, risk)
PV vs Discount Rate	Inverse relationship — higher discount rate means lower present value
Annuity	A series of equal periodic payments — ordinary annuity (end of period) vs annuity due (start of period)
Bond valuation	Bond’s fair value = PV of coupon

	payments + PV of face value at maturity
Bond price vs interest rate	Inverse relationship; also depends on coupon rate vs market rate (par/premium/discount)

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Unit II: Capital Budgeting

(BBA Sem 3 — F010301TN — 7 Lectures)

1. Meaning of Capital Budgeting

Capital Budgeting refers to the **process of planning and evaluating long-term investment decisions**, where a firm decides whether a particular long-term project or investment (e.g., purchasing new machinery, launching a new product line, expanding into a new facility) is worth undertaking.

These decisions are called “capital” budgeting decisions because they typically involve large sums of money committed for a long period of time, and once made, they are often difficult or costly to reverse. Capital budgeting essentially forms the practical application of the **Investment Decision** discussed in Unit I.

2. Objectives of Capital Budgeting

- **Selecting the most profitable investment projects** — Ensuring that funds are allocated to projects that maximize shareholder wealth (linking back to the objective of financial management discussed in Unit I).
 - **Efficient allocation of scarce capital resources** — Since a firm typically doesn’t have unlimited funds, capital budgeting helps prioritize among competing investment opportunities.
 - **Minimizing investment risk** — Systematic evaluation helps identify and account for the risks associated with long-term projects before committing significant resources.
 - **Long-term growth and survival** — Well-evaluated capital investments are essential for a firm’s long-term competitiveness, expansion, and survival in the market.
 - **Controlling capital expenditure** — Provides a structured framework to monitor whether actual investment outcomes match what was originally planned/expected.
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3. Process of Capital Budgeting

Capital budgeting is typically carried out through a structured, sequential process:

1. **Identification/Generation of Investment Proposals** — Ideas for potential projects are generated from various sources — production department (new equipment needs), marketing department (new product opportunities), or top management (strategic expansion plans).
2. **Screening and Evaluation of Proposals** — Each proposal is evaluated in terms of its technical feasibility, expected costs, expected cash flows/benefits, and risk, using various capital budgeting techniques (covered in detail below).
3. **Estimating Cash Flows** — Careful estimation of the initial investment outlay and the expected future cash inflows the project will generate over its useful life — this is one of the most critical steps, since the accuracy of all subsequent analysis depends on realistic cash flow estimates.
4. **Selection of the Project** — Based on the evaluation using capital budgeting techniques (Payback Period, NPV, IRR, Profitability Index), and considering the firm's available funds and strategic priorities, the most suitable project(s) are selected.
5. **Implementation** — Once approved, funds are allocated, and the project is executed — this involves converting the approved investment proposal into an actual functioning asset/operation.
6. **Performance Review (Post-Completion Audit)** — After implementation, actual performance (costs and returns generated) is compared against the original projections, to evaluate whether the investment decision was sound and to draw lessons for future capital budgeting decisions.

Exam tip: “Explain the process of capital budgeting” is a common 6-mark question — write the 6 steps above in sequence, similar to how you'd structure the HRP process or Marketing Research process in other subjects; examiners consistently reward clear step-by-step structuring for “process” questions.

4. Techniques of Capital Budgeting Decision

This is the single most important and most heavily tested section of this unit. Capital budgeting techniques are broadly divided into **Non-Discounting (Traditional) Methods** and **Discounting (Time-Adjusted) Methods**.

A. Payback Period Method (Non-Discounting Method)

Concept: The Payback Period is the **length of time required for a project's cumulative cash inflows to equal (recover) the initial investment outlay**. It essentially answers: "How quickly will I get my invested money back?"

Calculation logic:

- If cash inflows are equal each year: $\text{Payback Period} = \frac{\text{Initial Investment}}{\text{Annual Cash Inflow}}$
- If cash inflows are unequal each year: Cash inflows are accumulated year by year until they equal the initial investment; the exact payback period is then interpolated within the year in which recovery is completed.

Decision Rule: A project is accepted if its payback period is less than or equal to a predetermined/target payback period set by management. When comparing multiple projects, the one with the **shorter** payback period is generally preferred.

Advantages:

- Simple to understand and easy to calculate
- Useful in situations of high liquidity preference or high-risk environments, since it emphasizes quick recovery of investment
- Acts as a rough measure of a project's risk (a shorter payback generally means lower risk of loss)

Limitations:

- **Ignores the time value of money** — Treats a rupee received in year 1 the same as a rupee received in year 5 (unless using the "discounted payback" variant)
- **Ignores cash flows occurring after the payback period** — Two projects could have the same payback period, but one might generate substantially higher cash flows afterward; the plain payback method would treat them as equally attractive
- Does not directly measure profitability, only the speed of capital recovery

B. Net Present Value (NPV) Method (Discounting Method)

Concept: NPV is the **difference between the present value of a project's cash inflows and the present value of its cash outflows (initial investment)**, discounted at the firm's required rate of return (cost of capital). It directly reflects the addition to shareholder wealth that the project is expected to generate.

Formula (conceptual):

$$NPV = (\text{Sum of Present Values of all future Cash Inflows}) -$$

Initial Investment

Decision Rule:

- If **NPV > 0** → Accept the project (it is expected to increase shareholder wealth)
- If **NPV < 0** → Reject the project (it would decrease shareholder wealth)
- If **NPV = 0** → The project earns exactly the required rate of return; decision may depend on other factors
- When comparing multiple mutually exclusive projects, the project with the **higher NPV** is generally preferred.

Advantages:

- **Considers the time value of money** — All cash flows are properly discounted, unlike the payback period method
- **Considers all cash flows over the project's entire life**, not just those up to the payback point
- Directly measures the expected increase in shareholder wealth, aligning perfectly with the objective of financial management (SWM) discussed in Unit I
- Considered the most theoretically sound capital budgeting technique

Limitations:

- Requires an accurate estimate of the discount rate (cost of capital), which can be difficult to determine precisely
- Expressed in absolute rupee terms, which can make it harder to compare projects of very different sizes (a large project may show a bigger NPV simply because of its scale)

C. Profitability Index (PI) Method (Discounting Method)

Concept: The Profitability Index (also called the Benefit-Cost Ratio) is the **ratio of the present value of future cash inflows to the initial investment**. It is essentially a relative (ratio-based) version of NPV, useful for comparing projects of different sizes.

Formula (conceptual):

$$PI = \text{Present Value of Cash Inflows} \div \text{Initial Investment}$$

Decision Rule:

- If **PI > 1** → Accept the project (present value of benefits exceeds the cost — equivalent to a positive NPV)

- If **PI < 1** → Reject the project
- If **PI = 1** → Same conclusion as NPV = 0
- When ranking multiple projects (especially under capital rationing, i.e., limited funds), the project(s) with the **higher PI** are generally preferred, since PI shows value created per rupee invested.

Advantages:

- Considers the time value of money (like NPV)
- Particularly useful when comparing projects of different sizes, or when funds are limited (capital rationing), since it measures relative profitability rather than absolute rupee value

Limitations:

- Like NPV, requires an accurate discount rate
- Can sometimes give a different ranking than NPV when project sizes differ significantly, which may occasionally cause conflicting decisions

D. Internal Rate of Return (IRR) Method (Discounting Method)

Concept: The Internal Rate of Return is the **discount rate at which the present value of a project's cash inflows exactly equals the present value of its cash outflows** — in other words, it's the discount rate at which the project's NPV becomes zero. IRR represents the project's own intrinsic/expected rate of return.

Decision Rule:

- If **IRR > Cost of Capital (required rate of return)** → Accept the project (the project earns more than what it costs to finance it)
- If **IRR < Cost of Capital** → Reject the project
- When comparing multiple projects, the one with the **higher IRR** is generally preferred (though this can occasionally conflict with NPV ranking in certain situations).

Advantages:

- Considers the time value of money
- Expressed as a percentage rate of return, which many managers find intuitive and easy to compare against the firm's cost of capital or against other investment opportunities
- Does not require the discount rate to be known in advance (it is calculated as an output, rather than needed as an input, unlike NPV)

Limitations:

- Can be complex to calculate by hand, often requiring trial-and-error or interpolation methods
- Can give misleading results in projects with non-conventional cash flows (e.g., cash outflows occurring at multiple points in time), potentially yielding multiple IRRs
- May sometimes conflict with NPV in ranking mutually exclusive projects of different scale or timing patterns — in such conflicts, NPV is generally considered the more reliable method, since it directly measures wealth creation in absolute terms

Comparison of Capital Budgeting Techniques

BASIS	PAYBACK PERIOD	NPV	PROFITABILITY INDEX	IRR
Time value of money	Ignored	Considered	Considered	Considered
Measures	Time to recover investment	Absolute wealth added	Relative profitability (ratio)	Rate of return (%)
Decision rule	Accept if $\leq$ target period	Accept if $NPV > 0$	Accept if $PI > 1$	Accept if $IRR >$ cost of capital
Best suited for	Quick liquidity/risk screening	Overall wealth maximization decisions	Comparing projects of different sizes	Comparing rate return across options
Key limitation	Ignores cash flows after payback	Requires accurate discount rate	Can conflict with NPV ranking sometimes	Can give multiple/misleading results in unusual cash flow patterns

Exam tip: “Explain the various techniques of capital budgeting” is one of the most predictable 6-mark (sometimes even higher weightage) questions in this subject — structure your answer using all four methods in the order above (Payback → NPV → PI → IRR), with the concept, decision rule, and one advantage/limitation for each. The comparison table is excellent for a quick “differentiate between NPV and IRR” or “differentiate between payback period and NPV” style 2-mark question as well.

Quick Revision Summary

TOPIC	ONE-LINE TAKEAWAY
Capital Budgeting meaning	Process of planning and evaluating long-term investment decisions
Objectives	Maximize wealth, allocate capital efficiently, minimize risk, support long-term growth
CB process	Generate proposals → screen/evaluate → estimate cash flows → select project → implement → post-audit
Payback Period	Time to recover initial investment; ignores time value of money and post-payback cash flows
NPV	PV of inflows – initial investment; accept if $NPV > 0$; most theoretically sound method
Profitability Index	$PV \text{ of inflows} \div \text{initial investment}$; accept if $PI > 1$; useful for comparing different-sized projects
IRR	Discount rate at which $NPV = 0$; accept if $IRR > \text{cost of capital}$
Best overall method	NPV — most reliable, especially when it conflicts with IRR

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Unit III: Capital Structure

(BBA Sem 3 — F010301TN — 7 Lectures)

1. Concept of Capital Structure

Capital Structure refers to the **mixture or proportion of different sources of long-term funds (primarily debt and equity) that a firm uses to finance its overall operations and growth**. It is the practical outcome of the **Financing Decision** discussed in Unit I — essentially, capital structure answers the question: “In what proportion should the firm raise funds through debt versus equity?”

A firm’s capital structure typically consists of:

- **Equity Share Capital** — Funds raised by issuing ownership shares to shareholders
- **Preference Share Capital** — A hybrid source carrying fixed dividend rights, senior to equity but junior to debt
- **Debt (Debentures/Long-term Loans)** — Borrowed funds carrying a fixed obligation to pay interest, regardless of the firm’s profitability
- **Retained Earnings** — Internally generated funds reinvested in the business rather than distributed as dividends

The goal of capital structure decisions is to arrive at an **optimal capital structure** — the mix of debt and equity that minimizes the firm’s overall cost of capital and thereby maximizes the market value of the firm (linking directly back to the Shareholder Wealth Maximization objective from Unit I).

2. Factors Determining Capital Structure

Several factors influence how a firm decides on its ideal mix of debt and equity:

- **Cost of Capital** — Since debt is generally cheaper than equity (partly due to the tax-deductibility of interest payments), firms often prefer some debt to lower their overall cost of capital — but only up to a point, beyond which the risk of excessive debt raises the overall cost.
- **Risk Consideration (Financial Risk)** — Higher use of debt increases fixed financial obligations (interest payments), which increases the risk of financial distress or insolvency if the firm’s

earnings are volatile or insufficient. Firms with stable, predictable earnings can typically afford higher debt levels than those with volatile earnings.

- **Control Considerations** — Raising funds through additional equity share issues can dilute the ownership and control of existing shareholders/promoters, whereas debt does not affect ownership control — firms concerned about maintaining control may prefer debt financing.
- **Flexibility** — A firm may want to maintain some unused borrowing capacity/flexibility to raise funds quickly in the future if needed — over-reliance on debt now can limit this future flexibility.
- **Cash Flow/Debt Servicing Capacity** — The firm must realistically assess whether its expected future cash flows will be sufficient to comfortably service (pay interest and principal on) debt obligations.
- **Nature of Industry and Business Risk** — Industries with stable, predictable demand (e.g., utilities) can typically sustain higher debt levels than industries with volatile or cyclical demand (e.g., technology startups).
- **Legal and Regulatory Requirements** — Statutory guidelines, lending norms, and regulatory restrictions (e.g., debt-equity ratio limits imposed by lenders or regulators) can constrain the capital structure choices available to a firm.
- **Market Conditions** — Prevailing conditions in the capital markets (e.g., a booming stock market may make equity issuance more attractive; low interest rates may make debt more attractive) also influence the financing mix chosen at a given point in time.

Exam tip: “Factors determining capital structure” is a common 6-mark question — aim to list and briefly explain at least 5-6 of the factors above (Cost of Capital, Risk, Control, Flexibility, Cash Flow, Nature of Industry are the most commonly expected points).

3. Leverage

Meaning

Leverage, in a financial context, refers to the **use of fixed-cost sources of funds or fixed operating costs to magnify the potential returns (and risks) to equity shareholders**. In simple terms, leverage is like using a lever to amplify a small effort into a

larger effect — here, fixed costs (whether operating or financial) amplify the effect of a change in sales/revenue on the firm's earnings.

Objectives and Uses of Leverage

- **Magnifying returns to shareholders** — When used favorably (i.e., when returns generated exceed the fixed cost of funds), leverage increases the earnings available to equity shareholders beyond what would be possible using only equity financing.
- **Analyzing risk-return trade-offs** — Leverage analysis helps management and investors understand how sensitive a firm's earnings are to changes in sales, helping assess business and financial risk.
- **Optimizing capital structure decisions** — Understanding leverage helps a firm decide the ideal proportion of fixed-cost funds (debt) to include in its capital structure.
- **Investment decision-making** — Investors use leverage analysis to judge the riskiness of a firm's earnings stream before investing.

Types of Leverage

A. Operating Leverage

Operating Leverage arises from the existence of **fixed operating costs** (e.g., rent, depreciation, fixed salaries) in a firm's cost structure. It measures the sensitivity of a firm's **EBIT (Earnings Before Interest and Tax)** to changes in sales volume.

- A firm with high fixed operating costs relative to variable costs has **high operating leverage** — meaning a small change in sales produces a proportionately larger change in EBIT.
- High operating leverage indicates high **business risk** — since fixed costs must be paid regardless of the sales level, a fall in sales can sharply reduce (or even wipe out) operating profits.

Example logic: A capital-intensive manufacturing firm (with high fixed costs of machinery/plant) has higher operating leverage than a firm using mostly variable-cost labor.

B. Financial Leverage

Financial Leverage arises from the existence of **fixed financial costs** (i.e., interest on debt and preference dividends) in a firm's capital structure. It measures the sensitivity of **EPS (Earnings Per Share)** to changes in EBIT.

- A firm that uses a larger proportion of debt (and/or preference

capital) in its capital structure has **high financial leverage** — meaning a small change in EBIT produces a proportionately larger change in EPS available to equity shareholders.

- High financial leverage indicates high **financial risk** — since interest must be paid regardless of the firm’s operating performance, a decline in EBIT can disproportionately hurt returns to equity shareholders (and in severe cases, threaten the firm’s solvency).
- Financial leverage can work both ways: it is **favorable** (magnifies EPS positively) when the firm’s return on investment exceeds the cost of debt, and **unfavorable** (magnifies losses) when it does not — this is sometimes called “trading on equity.”

C. Combined (Total) Leverage

Combined Leverage is the **combined effect of both operating leverage and financial leverage**, measuring the overall sensitivity of a firm’s EPS to a change in sales. It essentially captures the firm’s total risk exposure — both business risk (from operating leverage) and financial risk (from financial leverage) — in one combined measure.

- Combined Leverage = Operating Leverage × Financial Leverage
- A firm with both high operating leverage and high financial leverage carries very high combined risk — even a small drop in sales can severely reduce EPS, since the effects of both types of leverage compound each other.

Comparison Table

BASIS	OPERATING LEVERAGE	FINANCIAL LEVERAGE	COMBINED LEVERAGE
Arises from	Fixed operating costs	Fixed financial costs (interest, pref. dividend)	Both fixed operating and financial costs
Measures sensitivity of	EBIT to change in Sales	EPS to change in EBIT	EPS to change in Sales
Type of risk indicated	Business risk	Financial risk	Total (overall) risk

Exam tip: “Explain operating, financial, and combined leverage” is a frequently asked 6-mark question — make sure you clearly state what each type of leverage measures (this comparison table captures the exact distinction examiners look for) and link operating leverage to business risk, financial leverage to financial risk.

4. Cost of Capital

Concept

Cost of Capital refers to the **minimum rate of return that a firm must earn on its investments to satisfy the expectations of its various sources of funds (investors and lenders)**. It represents the “price” a firm pays for using funds from different sources, and it serves as the discount rate used in capital budgeting techniques like NPV (as discussed in Unit II).

Since a firm typically raises funds from multiple sources (equity, preference shares, debt), each source has its own individual cost, which must eventually be combined into an overall cost of capital for the firm.

Cost of Debt

The Cost of Debt is the **effective rate of return that lenders/debt-holders require, adjusted for the tax benefit the firm receives on interest payments** (since interest is a tax-deductible expense, the effective/after-tax cost of debt is lower than the stated interest rate).

Key point: Because interest expense reduces taxable income, the **after-tax cost of debt is always lower than the before-tax cost of debt** — this tax advantage is one of the key reasons debt is often considered a cheaper source of finance compared to equity.

Cost of Preference Shares

The Cost of Preference Shares is the **rate of return required by preference shareholders**, based on the fixed dividend rate they are promised. Unlike interest on debt, preference dividends are **not tax-deductible** (they are paid out of after-tax profits), meaning there is no tax shield benefit on preference share capital, unlike debt.

Cost of Equity Shares

The Cost of Equity is the **rate of return that equity shareholders expect/require in exchange for bearing the risk of investing in the firm's ownership**. Equity is generally considered the most expensive source of capital, because:

- Equity shareholders bear the highest risk (they are paid only after all other claims — debt-holders and preference shareholders — are settled)
- There is no tax benefit on dividends paid to equity shareholders (paid out of after-tax profits, similar to preference dividends)
- Equity investors expect a higher return to compensate for this higher risk (higher risk demands higher expected return — a

fundamental principle in finance)

Weighted Average Cost of Capital (WACC)

WACC is the **overall (composite) cost of capital for the firm, calculated by weighting the cost of each individual source of finance (debt, preference shares, equity) by its proportion in the firm's total capital structure.**

Conceptual logic:

$$WACC = (Weight\ of\ Debt \times Cost\ of\ Debt) + (Weight\ of\ Preference\ Capital \times Cost\ of\ Preference\ Capital) + (Weight\ of\ Equity \times Cost\ of\ Equity)$$

Importance of WACC:

- Serves as the **discount rate** used in capital budgeting decisions (e.g., calculating NPV, as covered in Unit II) — a project should only be accepted if its expected return exceeds the firm's WACC.
- Acts as a benchmark to evaluate whether the firm's overall capital structure and investment decisions are creating or destroying shareholder value.
- Helps in deciding the optimal capital structure — since the goal is to find the debt-equity mix that **minimizes** the firm's WACC (and thereby maximizes firm value).

Exam tip: “Explain the cost of capital and its components” and “explain WACC and its significance” are both frequently tested 6-mark questions — make sure you understand the general logic that debt < preference shares < equity in terms of cost (from cheapest to most expensive), primarily due to the tax-deductibility of interest and the differing risk levels of each source.

Quick Revision Summary

TOPIC	ONE-LINE TAKEAWAY
Capital Structure	The mix of debt and equity used to finance the firm; goal is to find the optimal mix
Factors affecting capital structure	Cost of capital, risk, control, flexibility, cash flow, industry nature, regulation, market conditions
Leverage meaning	Use of fixed costs to magnify returns (and risk) to equity shareholders
Operating Leverage	Fixed operating costs; measures

	EBIT sensitivity to sales; reflects business risk
Financial Leverage	Fixed financial costs; measures EPS sensitivity to EBIT; reflects financial risk
Combined Leverage	Operating $\times$ Financial Leverage; measures EPS sensitivity to sales; reflects total risk
Cost of Debt	After-tax cost is lower than before-tax cost, due to tax-deductibility of interest
Cost of Preference Shares	Fixed dividend rate; no tax benefit (unlike debt)
Cost of Equity	Highest cost source, since equity holders bear the most risk; no tax benefit
WACC	Weighted average of costs of all sources of capital; used as the discount rate in capital budgeting

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Unit IV: Dividend Decision

(BBA Sem 3 — F010301TN — 6 Lectures)

1. Concept of Dividend Decision

The Dividend Decision is the third major decision in financial management (alongside Investment and Financing Decisions, covered in Unit I). It relates to **how much of the firm's earnings should be distributed to shareholders as dividends, and how much should be retained within the business for reinvestment.**

Dividend refers to the **portion of a company's after-tax profit distributed to its shareholders**, usually in proportion to the number of shares they hold. This decision is closely tied to the financing decision, because retained earnings (the portion of profit *not* paid out as dividends) represent one of the cheapest and most convenient internal sources of financing available to a firm.

2. Significance of Dividend Policy

Dividend Policy refers to the **guidelines and principles a company follows in deciding how much profit to distribute as dividends and how much to retain**, on a consistent basis over time.

Why dividend policy is significant:

- **Signals firm performance and stability** — A consistent or growing dividend often signals financial health and management's confidence in future earnings, which can positively influence investor perception and share price.
- **Affects shareholder wealth** — Since one of the core objectives of financial management is shareholder wealth maximization (as discussed in Unit I), the dividend decision directly affects how that wealth is realized — as immediate cash income (dividends) vs. potential future capital gains (from reinvested/retained earnings).
- **Influences the firm's capital structure and growth** — A firm that retains more earnings has more cheap internal capital available for growth and expansion, reducing dependency on external financing (debt or new equity issues).
- **Impacts investor clientele** — Different types of investors have

different preferences (e.g., retirees may prefer steady dividend income, while growth-oriented investors may prefer reinvestment and capital appreciation) — a firm's dividend policy can attract a specific "clientele" of investors aligned with that policy.

- **Affects market price of shares** — Dividend announcements/changes often trigger immediate market reactions, since they carry informational value about the firm's expected future performance.

3. Factors Affecting Dividend Decision

- **Profitability and Earnings Stability** — A firm with stable, predictable profits can more comfortably commit to a consistent dividend payout than a firm with volatile or unpredictable earnings.
- **Liquidity Position (Cash Availability)** — Even a profitable firm may not have sufficient cash on hand to pay dividends if its profits are tied up in receivables, inventory, or fixed assets — dividend decisions must consider actual cash availability, not just accounting profit.
- **Growth and Investment Opportunities** — A firm with abundant profitable investment opportunities may prefer to retain more earnings (pay lower dividends) to fund this growth internally, rather than distributing cash and later raising external funds (which is often costlier).
- **Legal and Contractual Restrictions** — Loan agreements or regulatory requirements sometimes place restrictions on the amount of dividends a firm can distribute (e.g., to protect lender interests).
- **Tax Considerations** — The relative tax treatment of dividends versus capital gains (for both the company and shareholders) can influence whether a firm prefers to pay higher dividends or retain earnings for reinvestment.
- **Shareholder Preferences/Expectations** — Existing shareholder expectations (built from the firm's dividend history) influence policy, since a sudden cut in dividends can be viewed very negatively by the market, even if a firm has valid strategic reasons for retaining more earnings.
- **Access to External Capital Markets** — A firm that can easily and cheaply raise external funds (through debt or new equity) may feel more comfortable paying out higher dividends, since it isn't as dependent on retained earnings to finance growth.
- **Inflation** — During periods of high inflation, firms may need to retain more earnings just to maintain the real (inflation-adjusted)

value and replacement cost of their existing assets.

Exam tip: “Factors affecting dividend decision” is a common 6-mark question — aim to list and briefly explain at least 5-6 factors from the list above (Profitability, Liquidity, Growth opportunities, Legal restrictions, and Shareholder preference are the most commonly expected core points).

4. Relevance Dividend Models

There are two schools of thought in dividend theory: the “relevance” school (which argues dividend policy *does* affect a firm’s value) and the “irrelevance” school (notably Modigliani-Miller, which argues it does *not*, under certain assumptions). Your syllabus specifically covers two important **relevance** models — Walter’s Model and Gordon’s Model — both of which argue that dividend policy *does* affect the market value of the firm.

A. Walter’s Model

Core Premise: James E. Walter argued that the choice of dividend policy almost always affects the value of the firm, and that this relationship depends on the **comparison between the firm’s internal rate of return on investments (r) and its cost of capital (k)**.

Key Logic:

- If a firm can reinvest retained earnings at a rate of return **higher** than its cost of capital ($r > k$), it should retain more earnings (pay lower dividends), since the firm can generate more wealth for shareholders by reinvesting than shareholders could earn on their own if the money were paid out as dividends.
- If a firm’s rate of return on investment is **lower** than its cost of capital ($r < k$), it should distribute more earnings as dividends (pay higher dividends), since shareholders could earn a better return by investing that money elsewhere themselves than the firm could generate internally.
- If $r = k$, the firm’s dividend policy has no effect on its share value — the firm is indifferent between retaining and distributing earnings.

Categorization of firms under Walter’s Model:

- **Growth firms** ($r > k$) — Should retain all earnings (pay zero/low dividends) to maximize share value
- **Declining firms** ($r < k$) — Should distribute all earnings as dividends (pay 100% dividend payout)

- **Normal firms** ($r = k$) — Dividend policy is irrelevant to share value; any payout ratio results in the same value

Key Assumptions of Walter's Model:

- All financing is done through retained earnings only (no external debt or equity financing)
- The firm's internal rate of return (r) and cost of capital (k) remain constant
- The firm has an infinite (perpetual) life

Exam tip: Walter's Model is a very commonly tested 6-mark question — the central point to remember is the **r vs. k comparison** and the resulting classification into growth, declining, and normal firms.

B. Gordon's Model (Dividend Growth Model)

Core Premise: Myron Gordon also argued that dividend policy affects a firm's value, but built his model specifically around the **"bird-in-hand" argument** — investors prefer current, certain dividend income over uncertain future capital gains from retained earnings, since the future is inherently riskier.

Key Logic:

- Because investors are risk-averse, they value near-term (certain) dividends more highly than distant (uncertain) future capital gains, even if the expected total return is mathematically the same.
- Investors therefore discount future capital gains at a **higher** rate than current dividend income (to account for this added uncertainty/risk).
- As a result, a firm that pays out higher dividends today (rather than retaining and promising future capital appreciation) will command a **higher market value**, according to Gordon's reasoning, since it satisfies the investor's preference for certainty.
- This is in direct contrast to the Modigliani-Miller irrelevance theory, which assumes investors are indifferent between dividends and capital gains under conditions of certainty — Gordon's model explicitly rejects this assumption by incorporating risk and uncertainty.

Key Assumptions of Gordon's Model:

- The firm is all-equity financed (no debt)
- No external financing is used — all investment is financed through retained earnings
- The firm's internal rate of return (r) and cost of capital (k) remain constant

- Retention ratio, once decided, remains constant forever
- Cost of capital (k) is greater than the growth rate (g), otherwise the valuation model becomes mathematically invalid

Comparison Table

BASIS	WALTER'S MODEL	GORDON'S MODEL
Core Argument	Dividend policy relevance depends on r vs. k	"Bird-in-hand" — investors prefer certain dividends over uncertain future capital gains
Key Variable Focus	Relationship between r (return on investment) and k (cost of capital)	Investor risk-aversion regarding the timing/certainty of returns
High dividend recommended when	$r < k$ (firm can't reinvest profitably)	Generally, regardless of r vs k — since investors always prefer certainty
Common ground	Both argue dividend policy is relevant and affects firm value (contrary to MM irrelevance theory)	Same

Exam tip: A common comparison question is "Differentiate between Walter's and Gordon's Dividend Models" — the key distinguishing point is that Walter's Model centers on the **r vs. k relationship**, while Gordon's Model centers on the **bird-in-hand (risk/certainty) argument**. Both conclude dividend policy is relevant, but for different underlying reasons.

5. Working Capital Management

Meaning

Working Capital refers to the **funds required to finance a firm's short-term, day-to-day operations** — essentially, the capital needed to manage current assets (like cash, inventory, and receivables) net of current liabilities (like accounts payable). It represents the operational, short-term counterpart to the long-term capital structure decisions covered in Unit III.

- **Gross Working Capital** — Refers to the firm's total investment in current assets (cash, inventory, receivables, etc.)
- **Net Working Capital** — Refers to the difference between current assets and current liabilities (Current Assets – Current

Liabilities); a positive net working capital indicates the firm has sufficient short-term assets to cover its short-term obligations.

Types of Working Capital

- **Permanent (Fixed) Working Capital** — The minimum level of current assets a firm must maintain at all times to continue its operations smoothly, regardless of seasonal fluctuations in business activity. This is a relatively stable, ongoing requirement.
- **Temporary (Variable/Fluctuating) Working Capital** — The additional working capital required to meet seasonal or temporary increases in business activity (e.g., extra inventory needed during a festive season rush). This fluctuates based on the specific demands of the business cycle at a given time.

Operating Cycle

The Operating Cycle (also called the Cash Conversion Cycle) refers to the **time period involved in converting raw materials/inventory into finished goods, then into sales, and finally into cash received from customers**. It effectively measures how long a firm's cash remains "tied up" in its day-to-day operations before it is converted back into usable cash.

Typical stages of the operating cycle:

1. Purchase of raw materials (on credit or cash)
2. Conversion of raw materials into work-in-progress, then finished goods
3. Sale of finished goods (often on credit, creating accounts receivable)
4. Collection of cash from debtors/customers

A **shorter operating cycle** means the firm converts its investments back into cash more quickly, requiring less working capital to sustain operations — this is generally seen as more efficient. A **longer operating cycle** ties up more funds for longer, increasing the firm's working capital requirement.

Determinants of Working Capital Requirement

- **Nature of Business** — Trading/service businesses generally require less working capital than manufacturing businesses (which need to fund raw materials, work-in-progress, and finished goods inventory).
- **Scale/Size of Operations** — Larger firms with higher production and sales volumes generally require more working capital in absolute terms.
- **Length of Operating Cycle** — A longer operating cycle (from raw material purchase to cash collection) increases working

capital requirements, as discussed above.

- **Credit Policy (Receivables and Payables)** — Liberal credit terms extended to customers increase the working capital tied up in receivables; conversely, favorable credit terms received from suppliers reduce the firm's own working capital burden.
- **Seasonality of Business** — Businesses with pronounced seasonal peaks (e.g., festival-driven retail) require higher temporary working capital during peak periods.
- **Growth and Expansion Plans** — Growing firms typically require increasing amounts of working capital to support higher levels of inventory, receivables, and operational activity.
- **Availability of Raw Materials** — If raw materials are not consistently available, firms may need to stock larger inventories as a buffer, increasing working capital needs.
- **Business/Industry Cycle Fluctuations** — During periods of economic boom, businesses often need more working capital to support higher sales activity; during a downturn, requirements typically decrease.

Exam tip: “Determinants of working capital requirement” and “explain the operating cycle” are both frequently tested 6-mark questions — for the operating cycle question, make sure to describe the sequential flow (raw material → WIP → finished goods → sale/receivables → cash) clearly, since examiners look for this logical sequence.

Quick Revision Summary

TOPIC	ONE-LINE TAKEAWAY
Dividend Decision	How much profit to distribute as dividends vs. retain for reinvestment
Dividend Policy significance	Signals firm health, affects shareholder wealth, influences capital structure and investor clientele
Factors affecting dividend decision	Profitability, liquidity, growth opportunities, legal restrictions, tax, shareholder preference
Walter's Model	Dividend relevance depends on r (return) vs k (cost of capital); growth firms retain, declining firms distribute
Gordon's Model	“Bird-in-hand” — investors prefer certain current dividends over

	uncertain future capital gains
Working Capital meaning	Funds needed to finance short-term, day-to-day operations
Gross vs Net Working Capital	Gross = total current assets; Net = current assets – current liabilities
Types of Working Capital	Permanent (stable minimum) vs Temporary (seasonal/fluctuating)
Operating Cycle	Time from raw material purchase to cash collection from sales; shorter = more efficient
Determinants of WC requirement	Nature of business, scale, operating cycle length, credit policy, seasonality, growth, raw material availability